

Exercises 2.2

3. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

a) $A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$ b) $A \cap B = \{3\}$ c) $A - B = \{1, 2, 4, 5\}$
d) $B - A = \{0, 6\}$

In Exercises 5-10 assume that A is a subset of some underlying universal set

U.

5. Prove the complementation law in Table 1 by showing that $\bar{\bar{A}} = A$.

Solution: Let $x \in \bar{A}$. By Defn. 5, $x \notin A$. By defn. of not belongs to operator, $\neg(x \in A)$ is True. By defn. of NOT, $\neg(\neg(x \in A))$ is False. By Double-negation Law of Propositions, $x \in A$ is False. $\therefore$ By Defn. 5, $x \notin \bar{\bar{A}}$ is False. By Defn. of not belongs to operator, $\neg(x \in \bar{\bar{A}})$ is False.